CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

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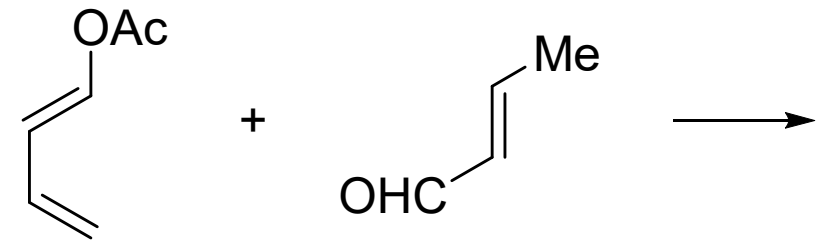
chmzhu@nus.edu.sg

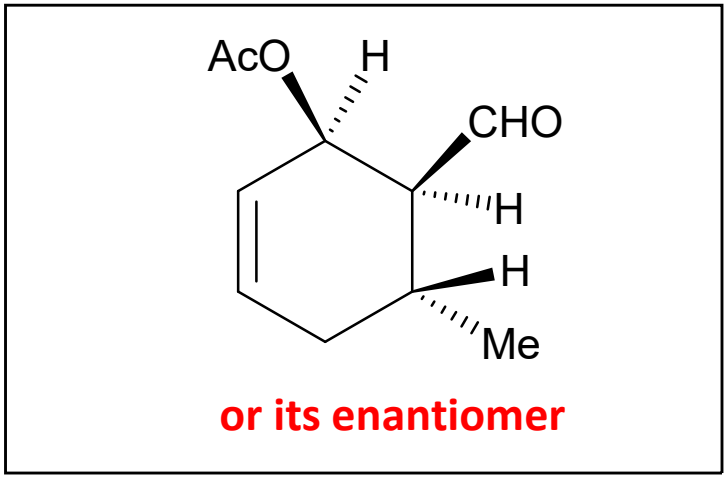
Cycloadditions (part 3)

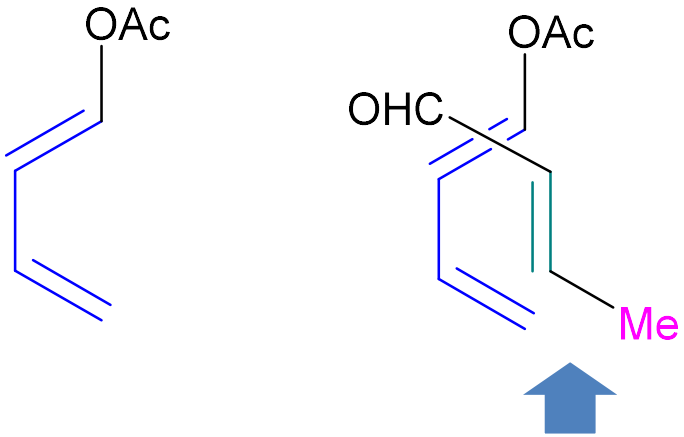
Overview

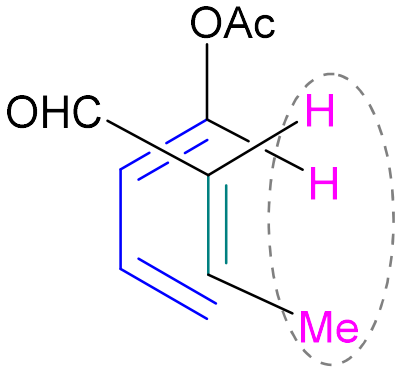
- Photochemical [2+2] cycloadditions
- Thermal [2 + 2] cycloadditions
  - Molecular orbitals and Woodward–Hoffmann rules
- [3 + 2] 1,3-dipolar cycloadditions
  - Nitrones
  - Nitrile oxide
  - Ozonolysis and its reaction mechanism
- Applications of cycloaddition reactions
  - Siloxydienes and Aminosiloxydienes
  - Bioconjugate Chemistry ([3+2] reactions of azides)

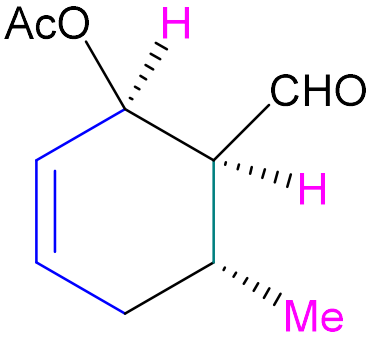
Practice





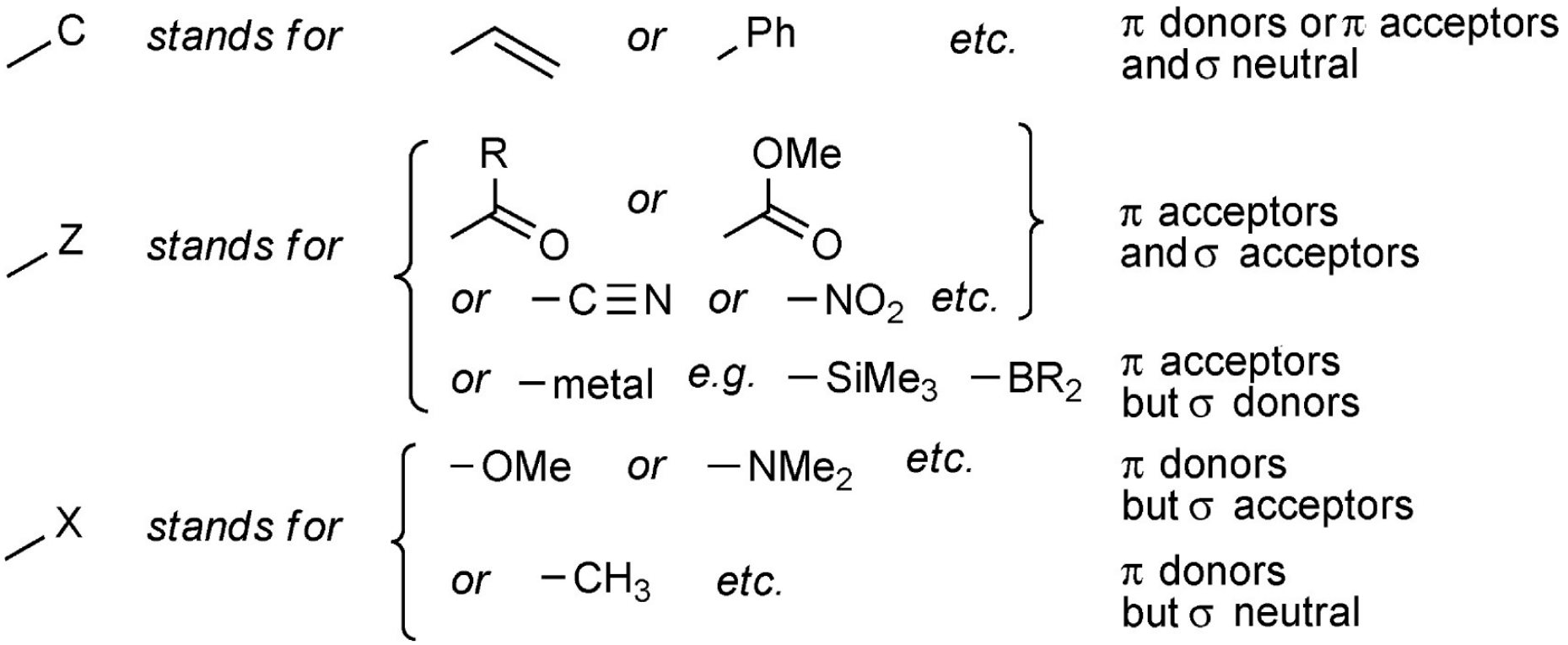




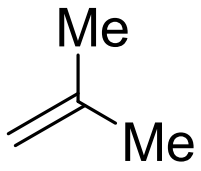


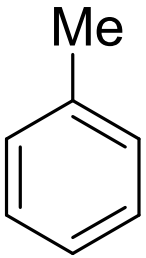
- Step 1: regioselectivtiy (ortho-/para-)
- Step 2: stereoselectivty (endo-)
- Step 3: stereospecificity

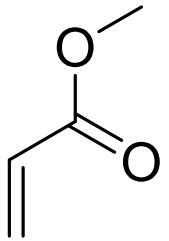
Substituents classifications

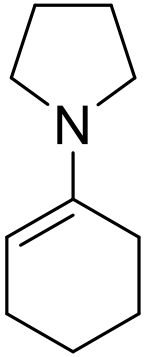


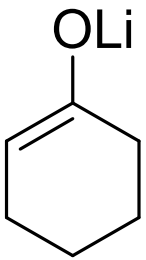
Definitions and character of C-, Z- and X- type substituents

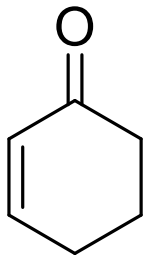

Me
Me

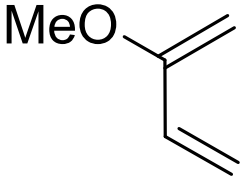

Me

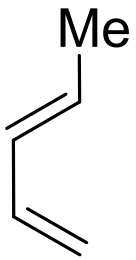

O
O

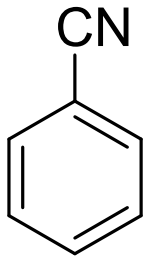

N


OLi

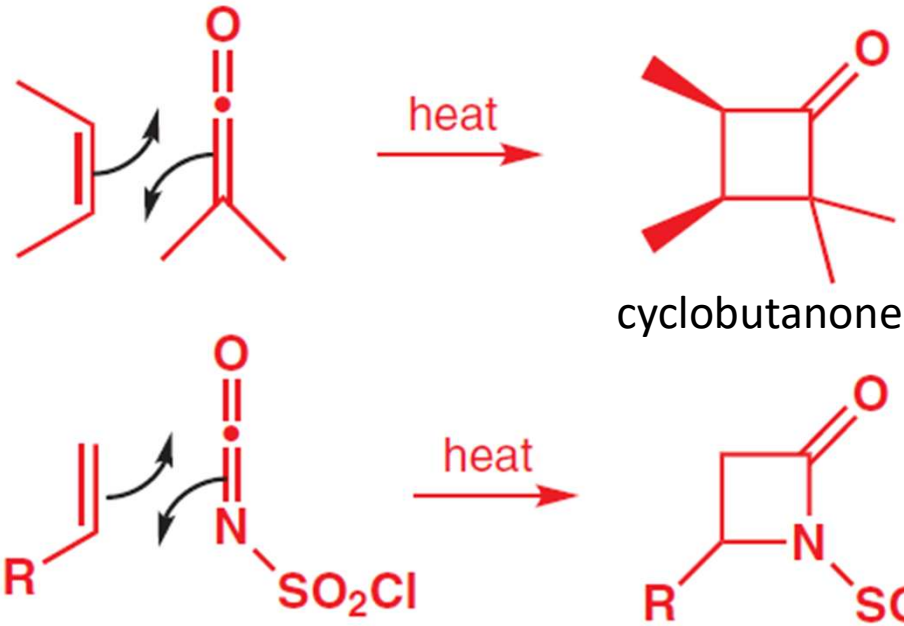

O


MeO


Me

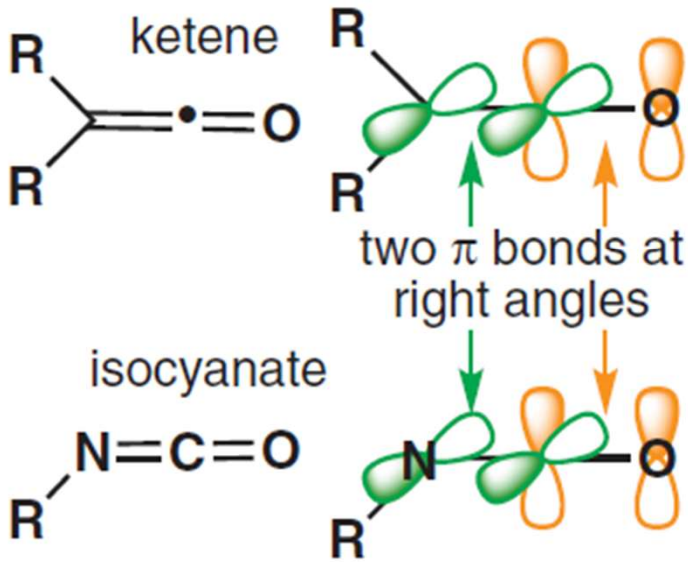

CN

14. Thermal [2 + 2] cycloadditions

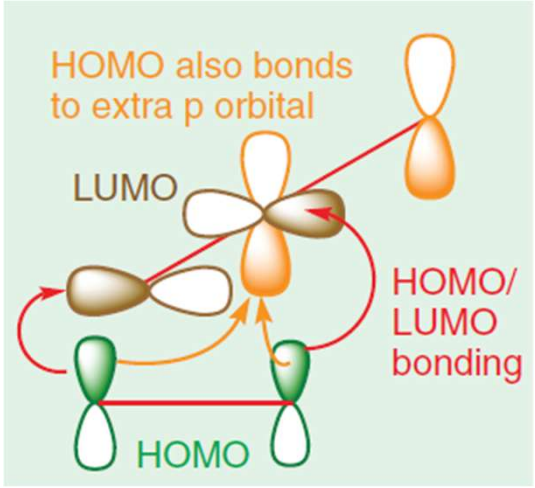


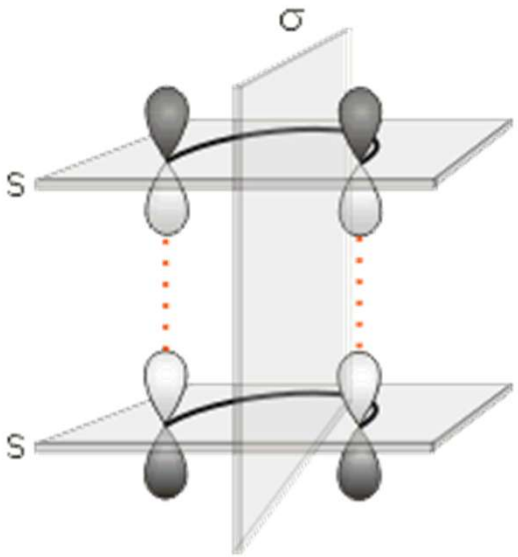
cyclobutanone

β-lactam

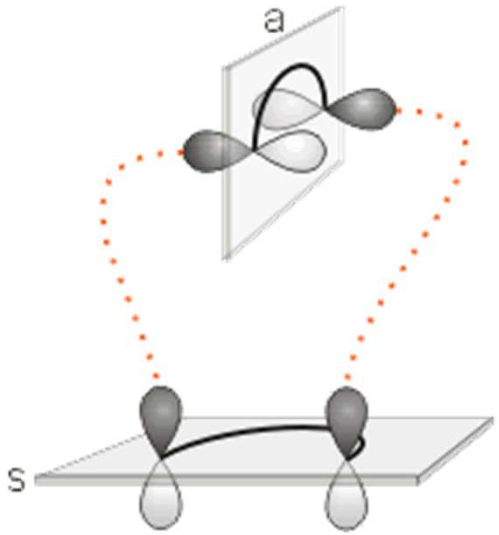


cumulated double bonds

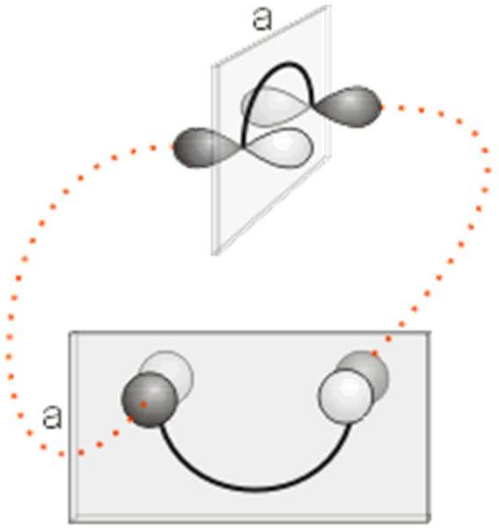




Supra-supra



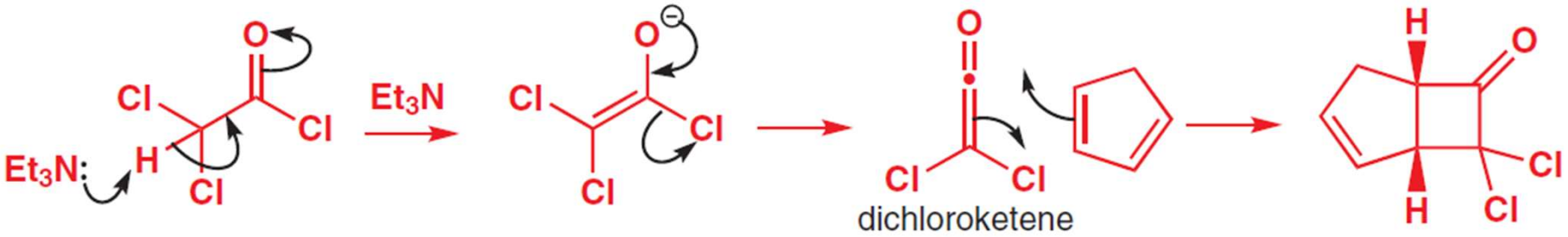
Supra-antara



Antara-antara

Thermal [2 + 2] cycloadditions

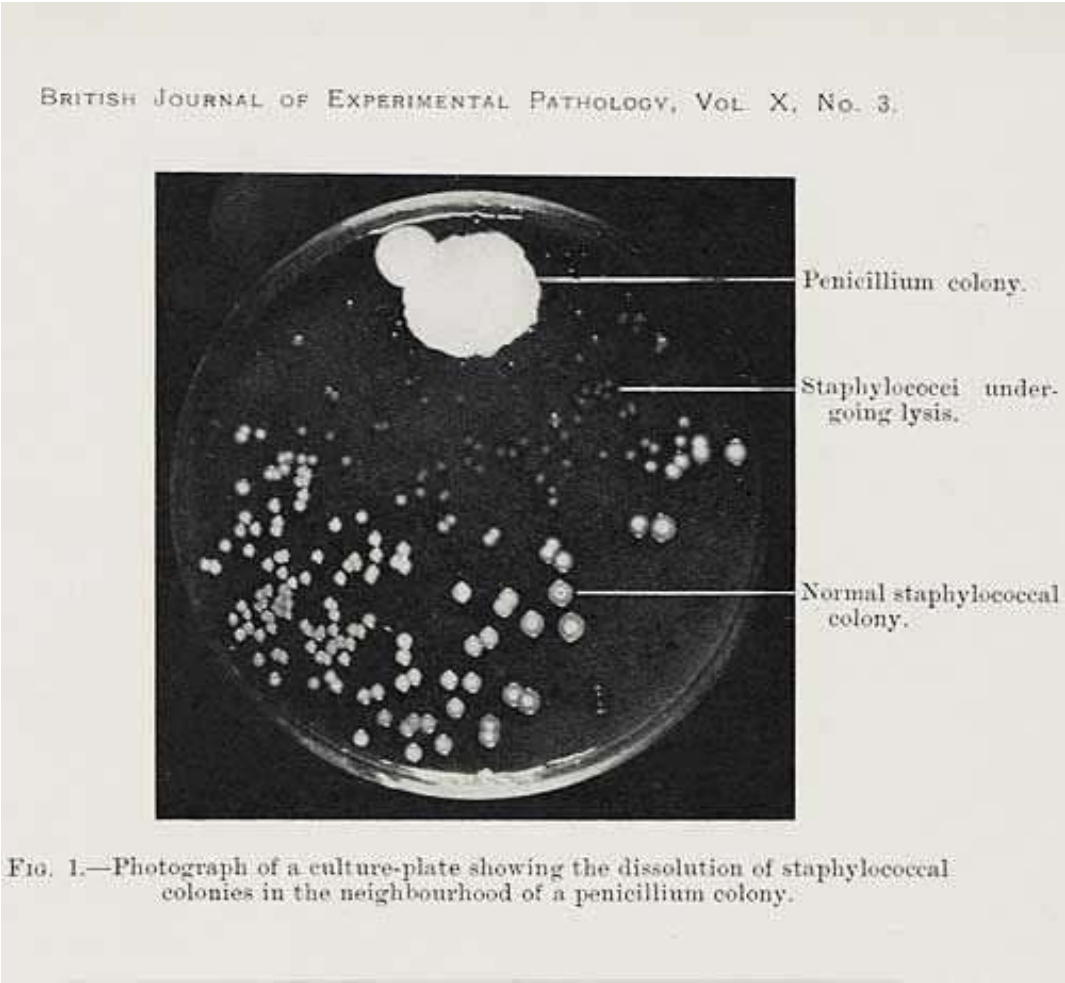
- cyclobutanone synthesis using ketene



Regioselectivity
(Hint: the carbonyl carbon is electrophilic)

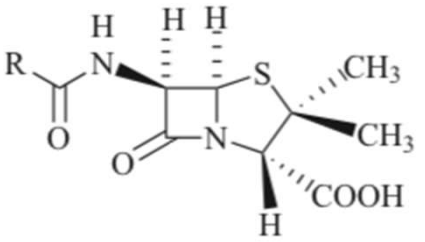
In this example, the [2 + 2] cycloaddition is faster than [4+2] cycloaddition

Penicillin as antibacterial agents





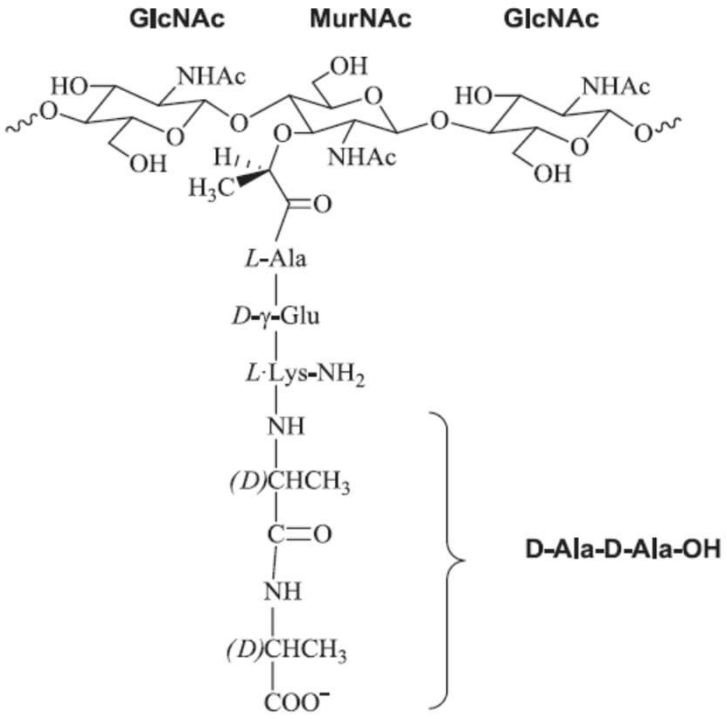
Alexander Fleming (Nobel Prize 1945)



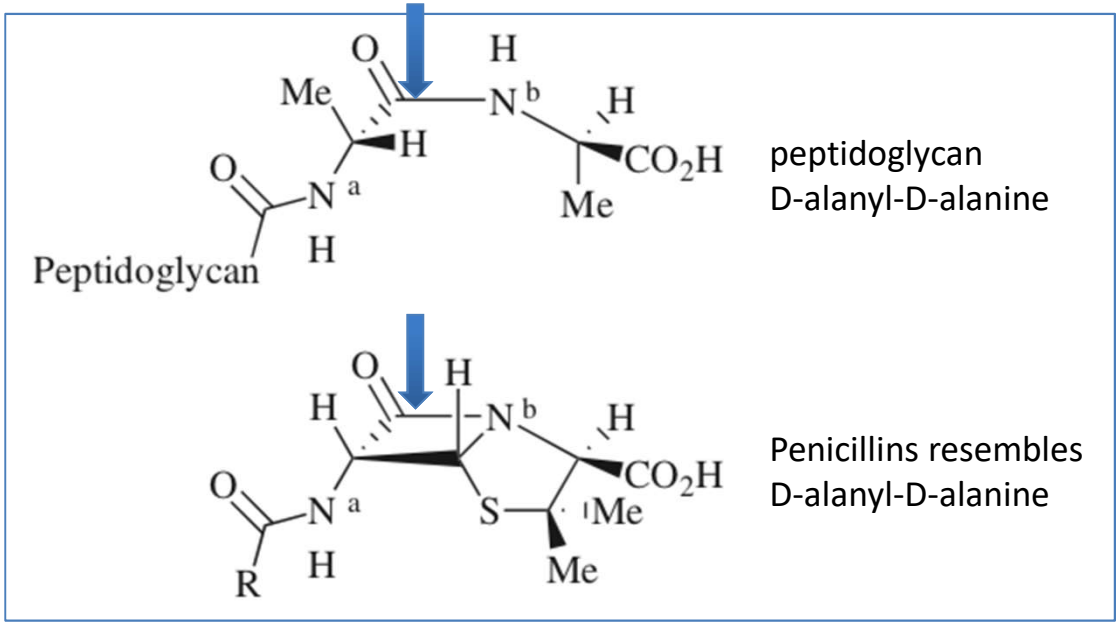
Penicillin V (R = $PhOCH_2$)
Penicillin G (R = CH_2Ph)

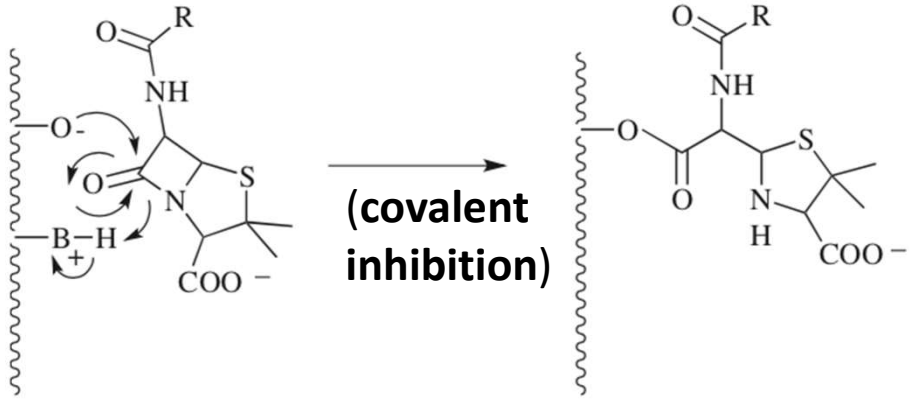
https://www.nobelprize.org/nobel_prizes/medicine/laureates/1945/fleming-lecture.html

Penicillin: mechanism of action



Structure of a peptidoglycan segment prior to cross-linking with another peptidoglycan fragment by transpeptidase

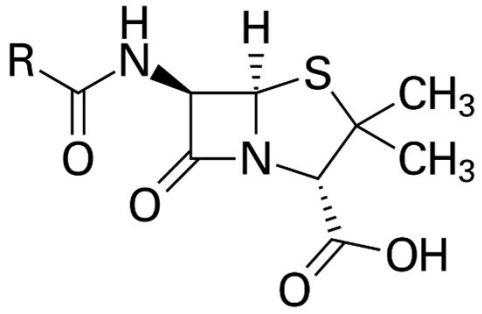




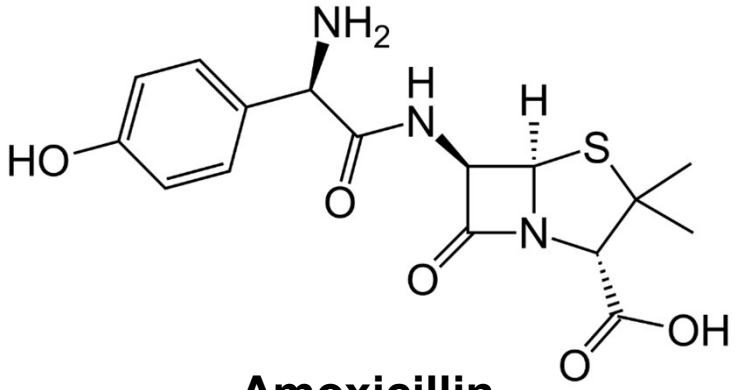
Acylation of peptidoglycan transpeptidase by Penicillins

Thermal [2 + 2] cycloadditions

- β-lactams synthesis



Penicillin



Amoxicillin

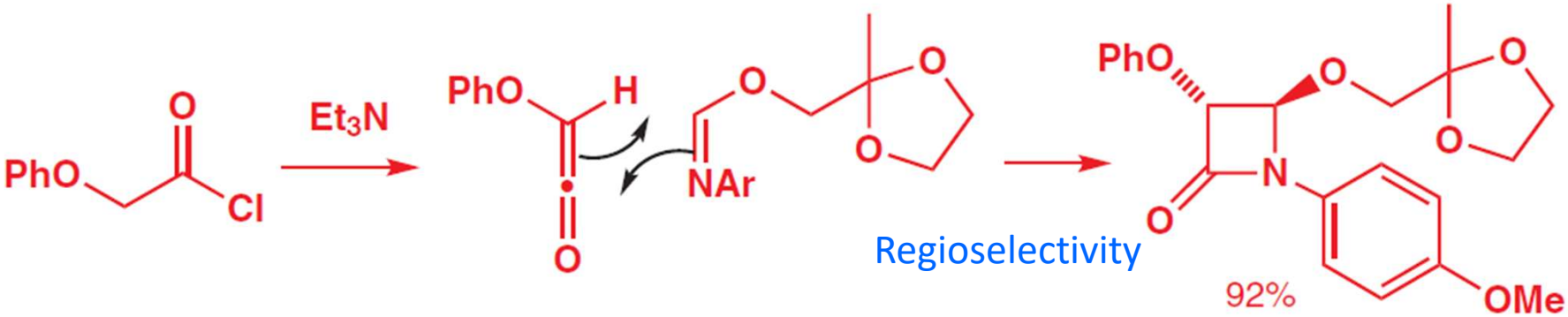
Two approaches for disconnection through [2 + 2] cycloaddition

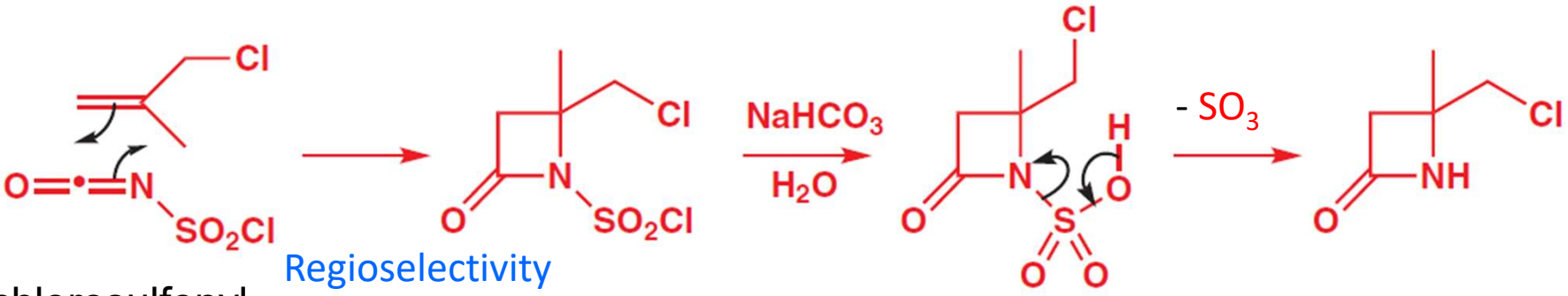




Thermal [2 + 2] cycloadditions

- β-lactams synthesis

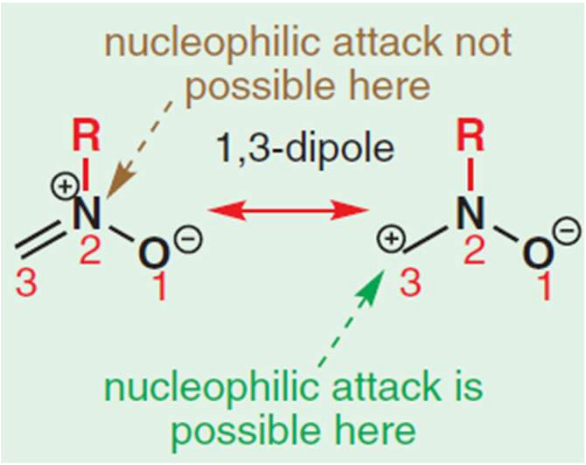




chlorosulfonyl isocyanate

15. 1,3-dipoles

1,3-dipoles are both electrophilic (on one end) and nucleophilic (on the other end).

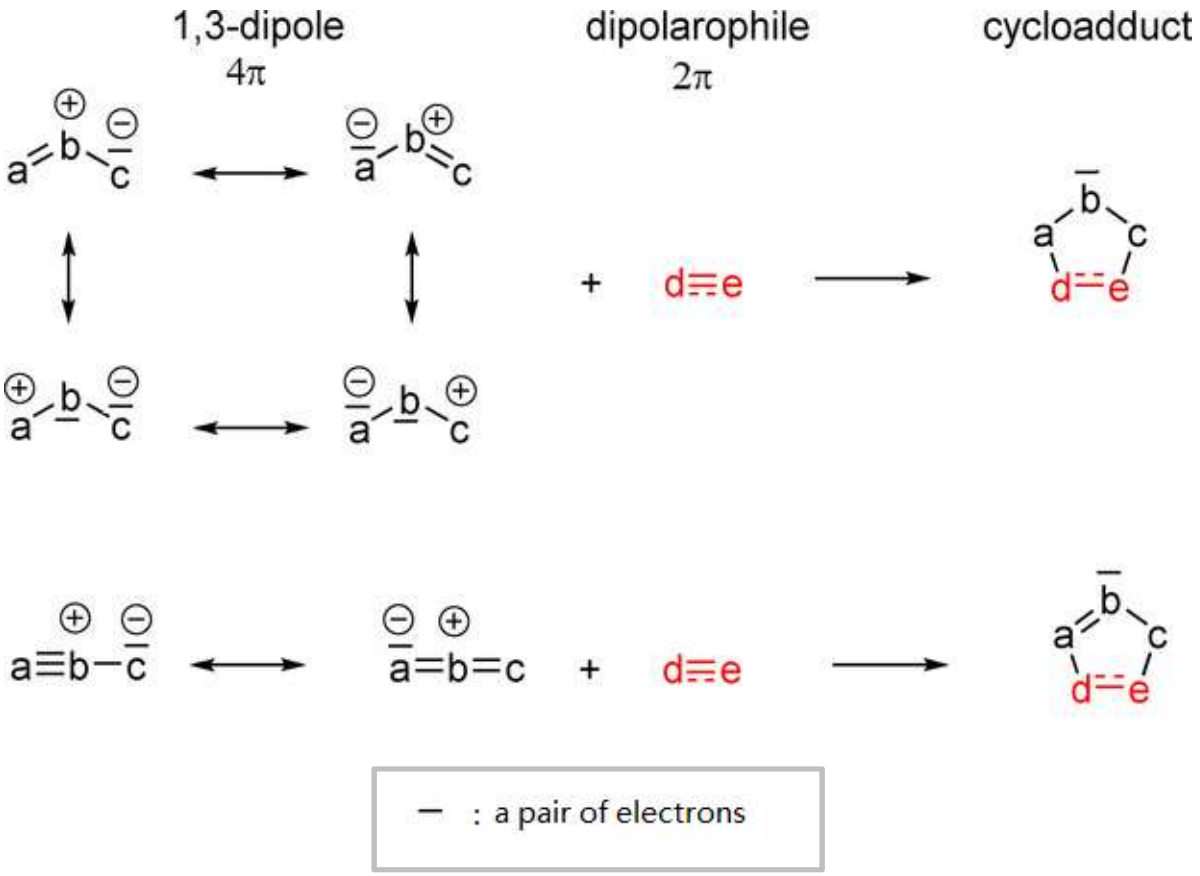


$:\overset{+}{N}{=}\ddot{N}{-}\bar{C}R_2$	↔	$:N{\equiv}\overset{+}{N}{-}\bar{C}R_2$	Diazoalkane
$:\overset{+}{N}{=}\ddot{N}{-}\bar{N}R$	↔	$:N{\equiv}\overset{+}{N}{-}\bar{N}R$	Azide
$R\overset{+}{C}{=}N{-}\bar{C}R_2$	↔	$RC{\equiv}\overset{+}{N}{-}\bar{C}R_2$	Nitrile ylide
$R\overset{+}{C}{=}\ddot{N}{-}\bar{N}R$	↔	$RC{\equiv}\overset{+}{N}{-}\bar{N}R$	Nitrile imine
$R\overset{+}{C}{=}N{-}\bar{O}:$	↔	$RC{\equiv}\overset{+}{N}{-}\bar{O}:$	Nitrile oxide
$R_2\overset{+}{C}{-}\ddot{N}{-}\bar{C}R_2$	↔	$R_2C{=}\overset{+}{N}(R){-}\bar{C}R_2$	Azomethine ylide
$R_2\overset{+}{C}{-}\ddot{N}(R){-}\bar{O}:$	↔	$R_2C{=}\overset{+}{N}(R){-}\bar{O}:$	Nitrone
$R_2\overset{+}{C}{-}\ddot{O}{-}\ddot{O}:^-$	↔	$R_2C{=}\overset{+}{O}{-}\ddot{O}:^-$	Carbonyl oxide

Count the number of electrons in the conjugated pi system of each compound.

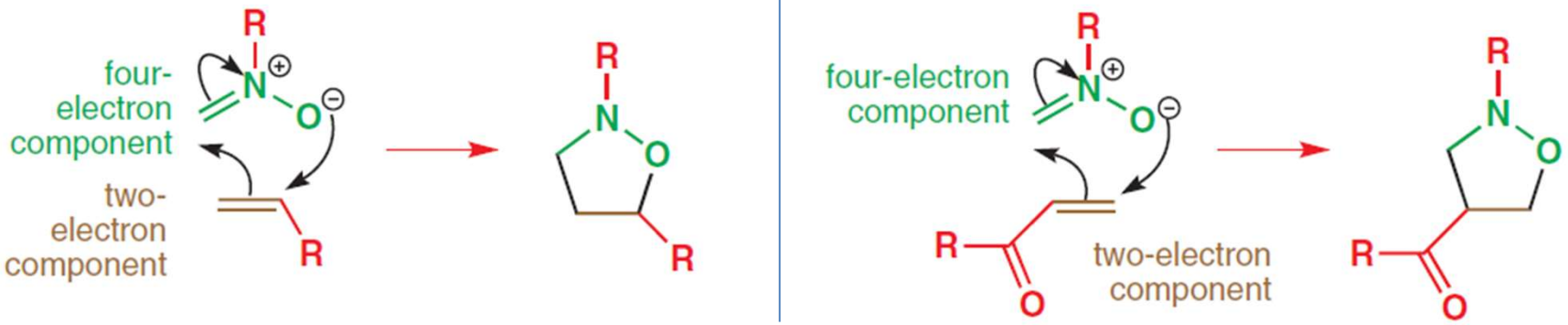
1,3-dipolar cycloadditions

- [3 + 2] cycloaddition between 1,3-dipole and dipolarophile.

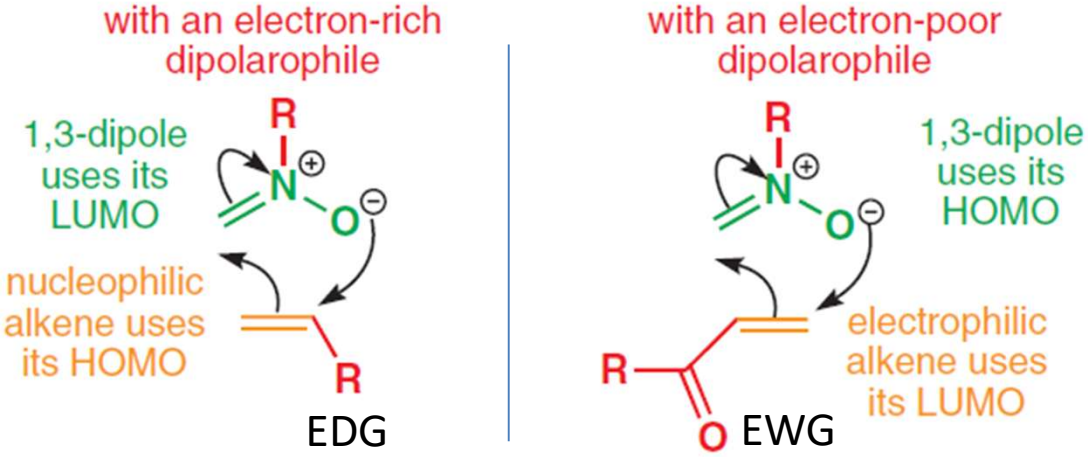


Nitrone as 1,3-dipole

- Regioselectivity: electronic effect (HOMO-LUMO interactions)

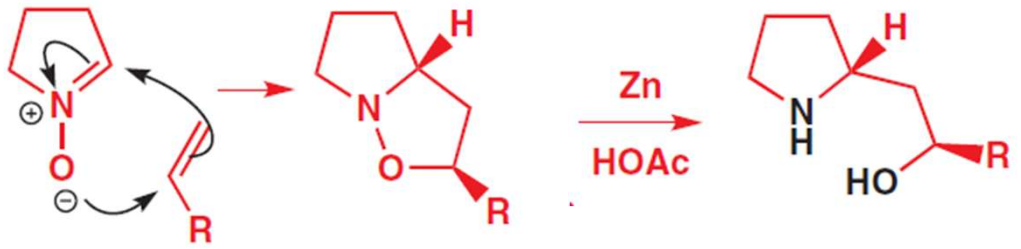


Nitrones are both electrophilic and nucleophilic

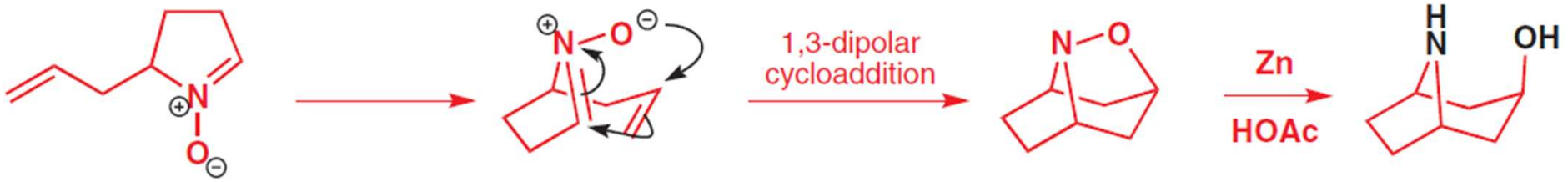


Nitrone as 1,3-dipole

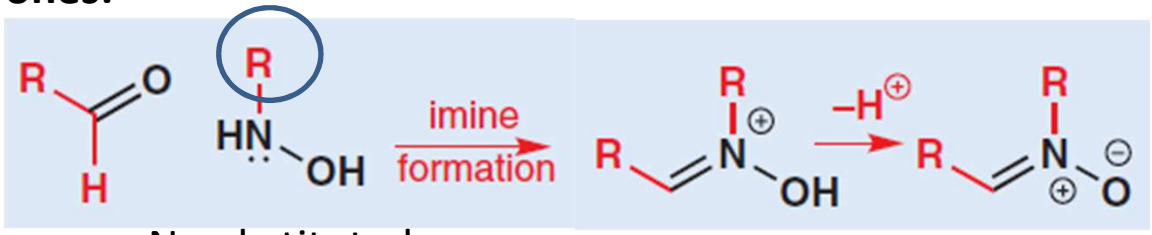
- Stereoselectivity: minimal steric hindrance (exo-transition state).
  - Cyclic nitrones



Upon reduction:
1,3-amino-alcohols

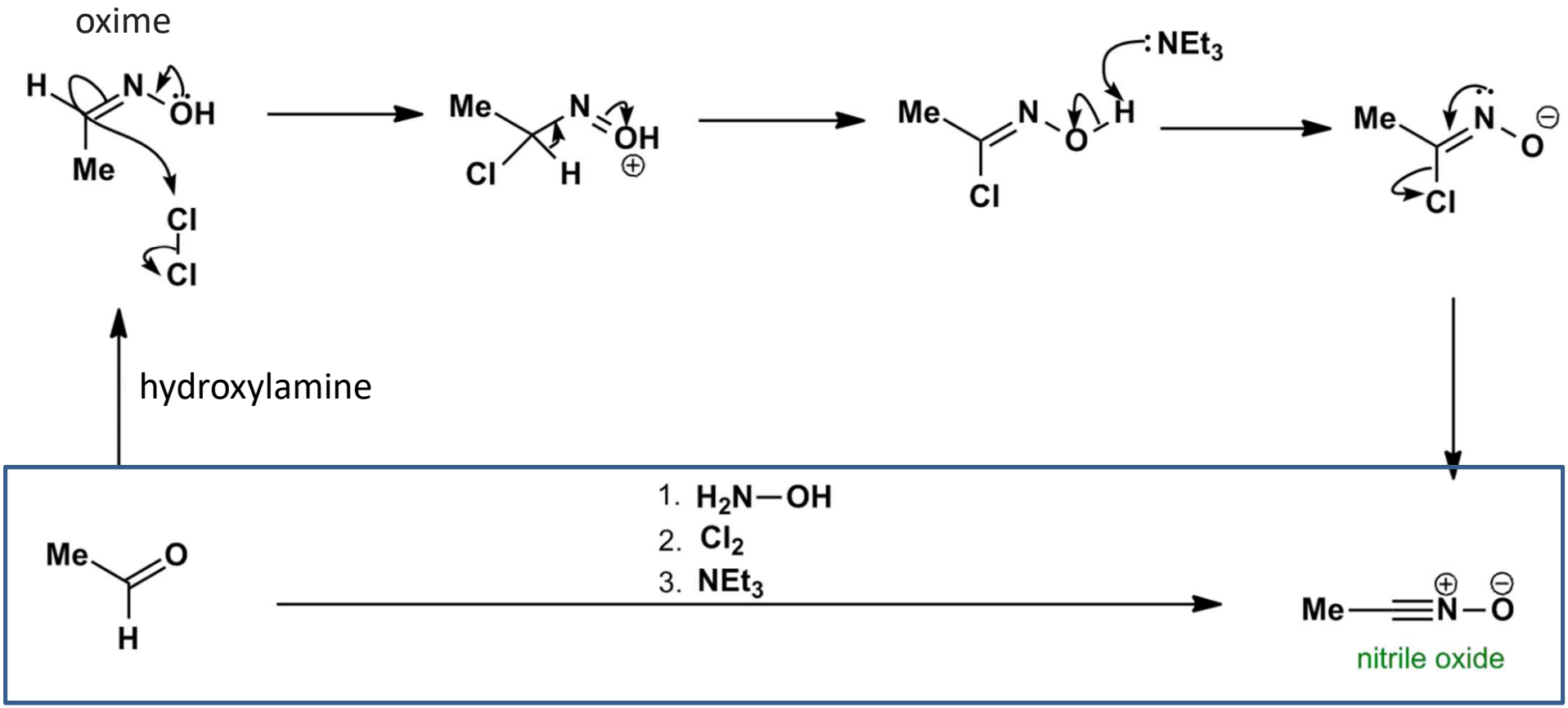


Preparation of nitrones:



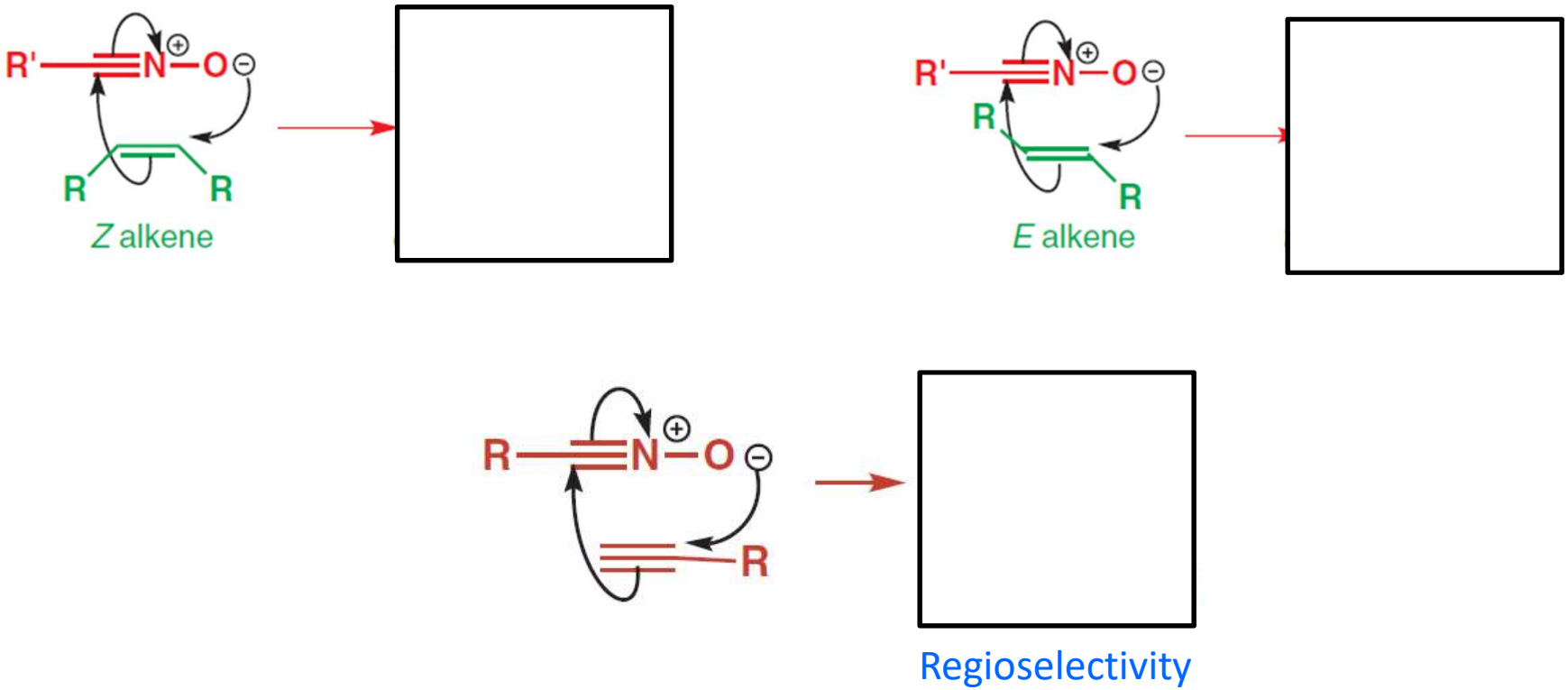
N-substituted
hydroxylamine

Preparation of nitrile oxide via reaction between oxime and chlorine



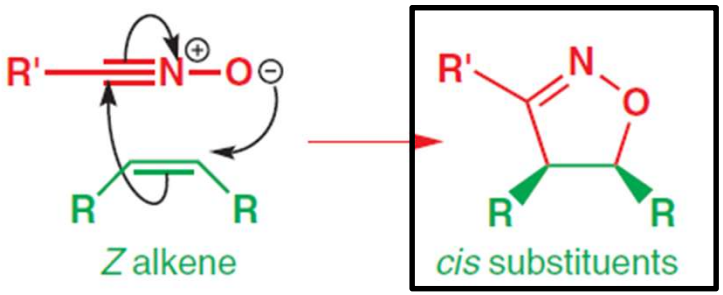
Nitrile oxide as 1,3-dipole

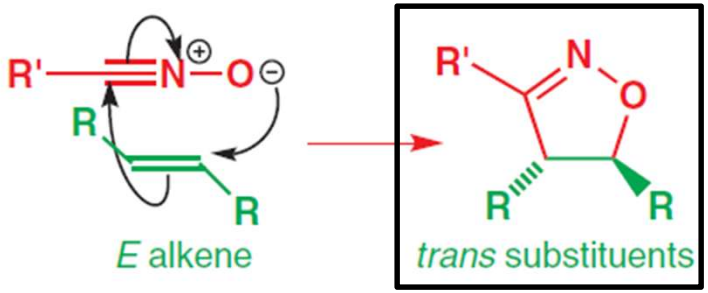
- Stereospecificity within substrate

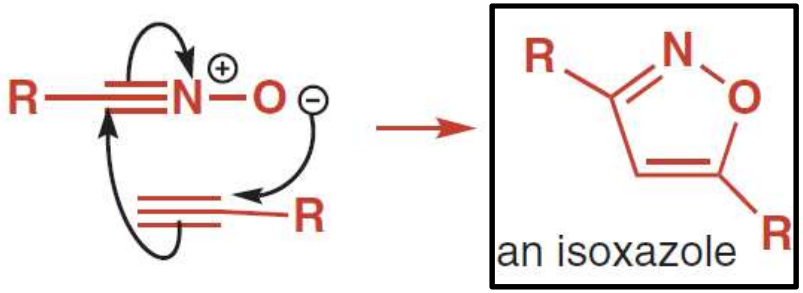


Nitrile oxide as 1,3-dipole

- Stereospecificity within substrate
  - Nitrile oxide

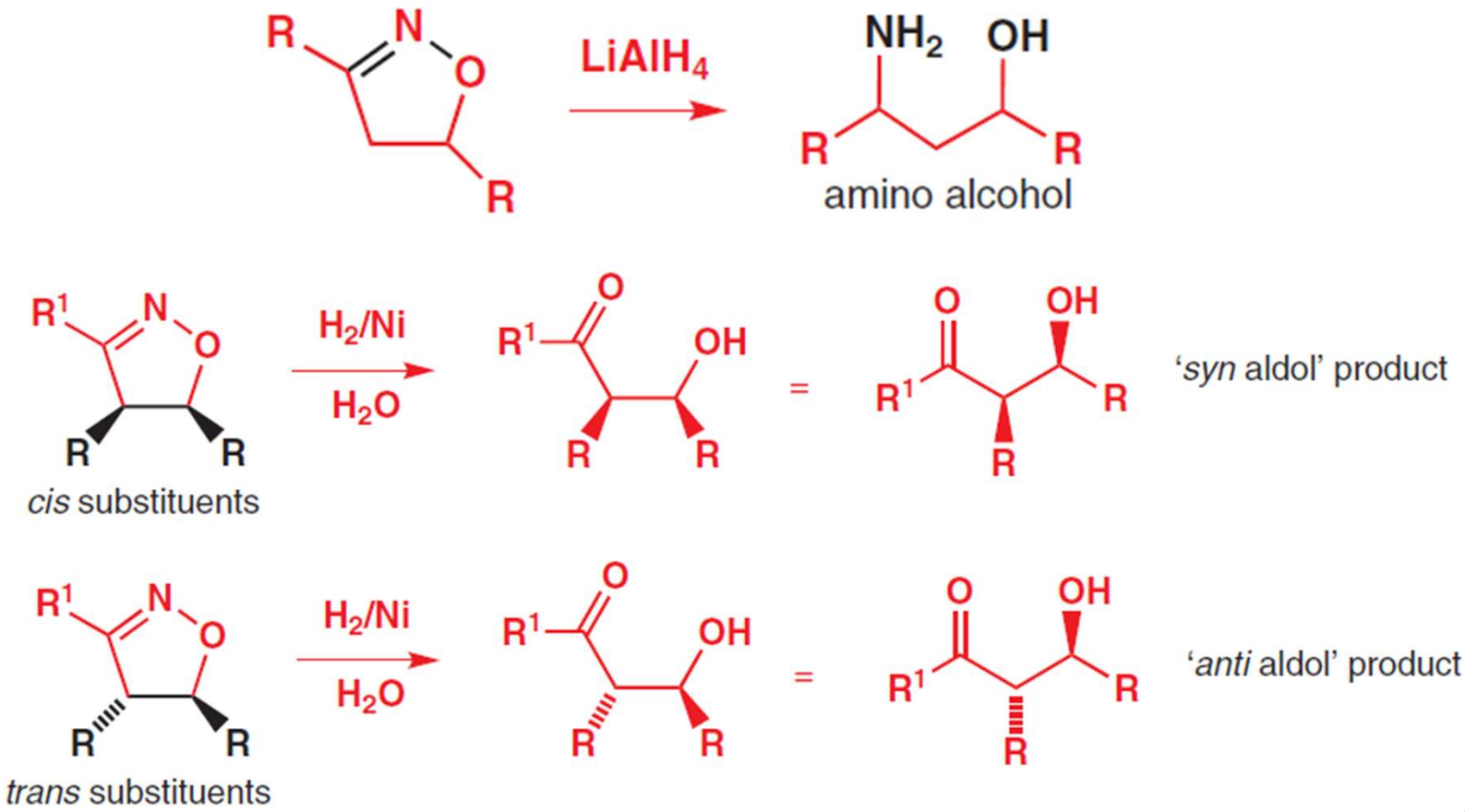






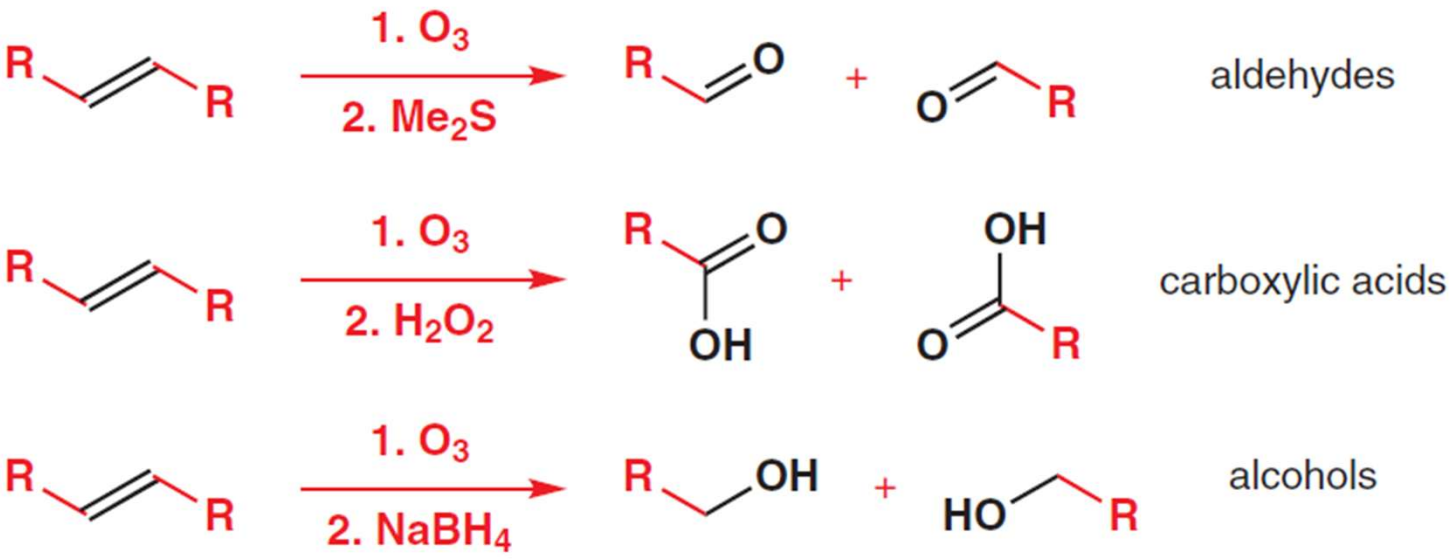
Nitrile oxide as 1,3-dipole

- Synthetic applications of products

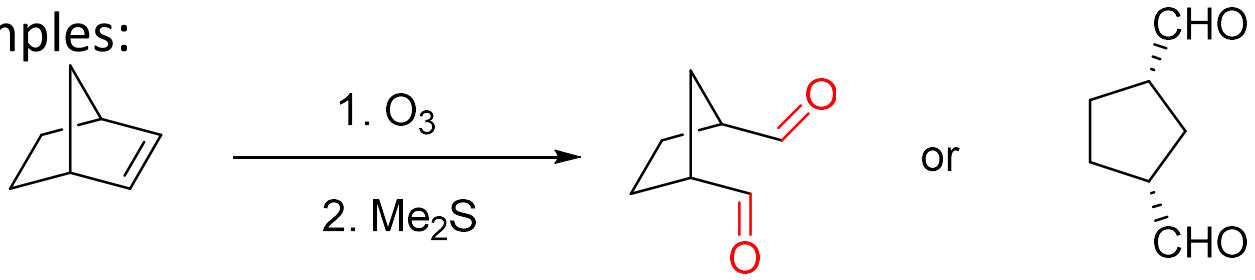


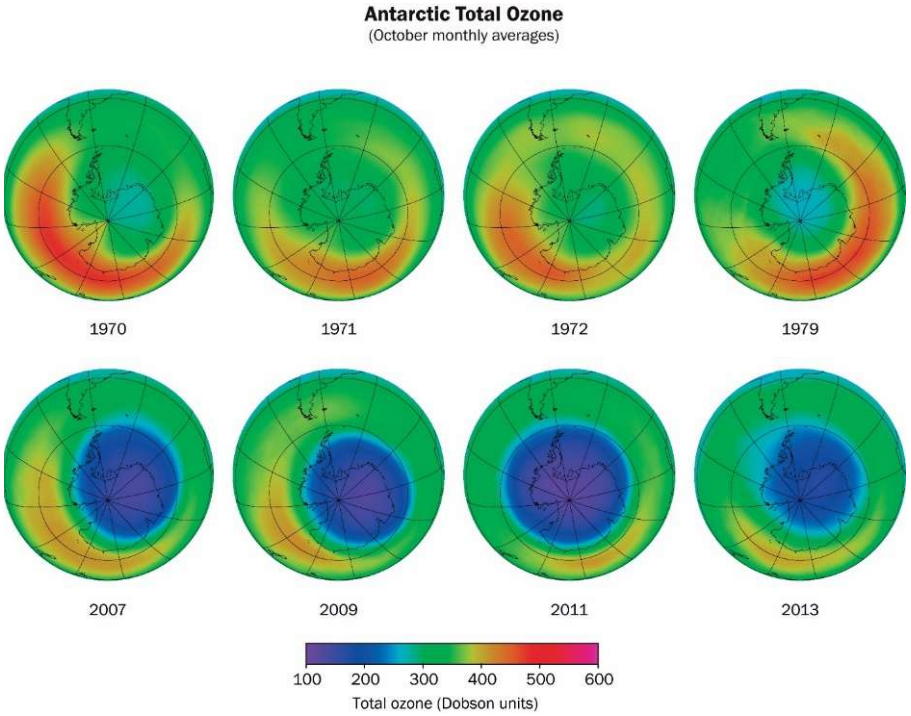
16. Ozonolysis

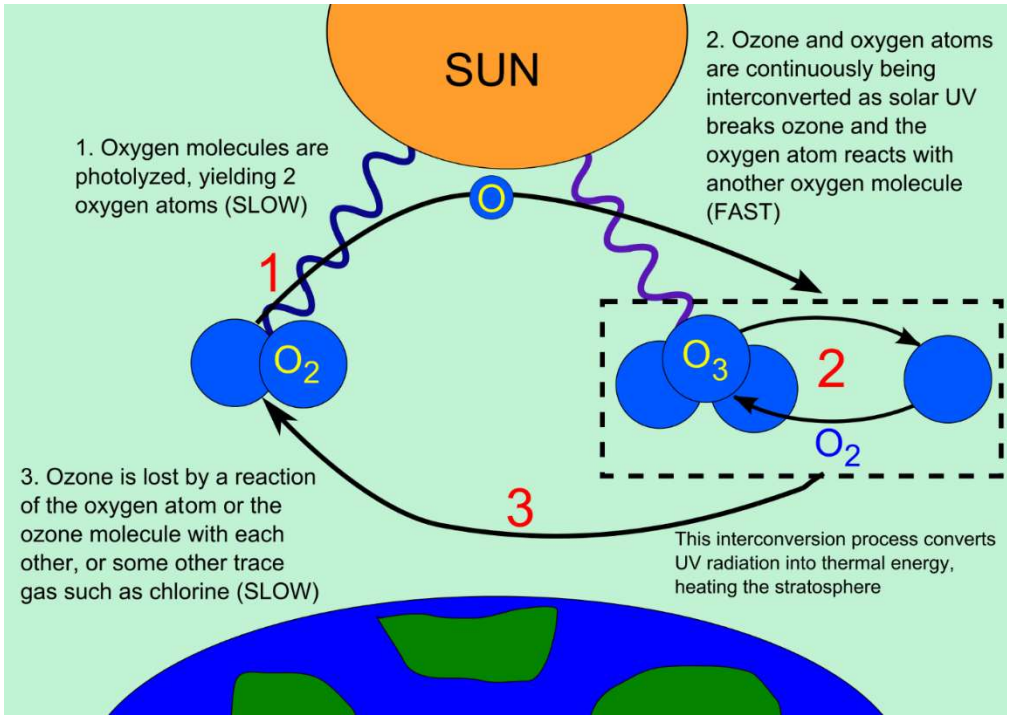
- Ozone cleaves double bonds to give ketones and aldehydes

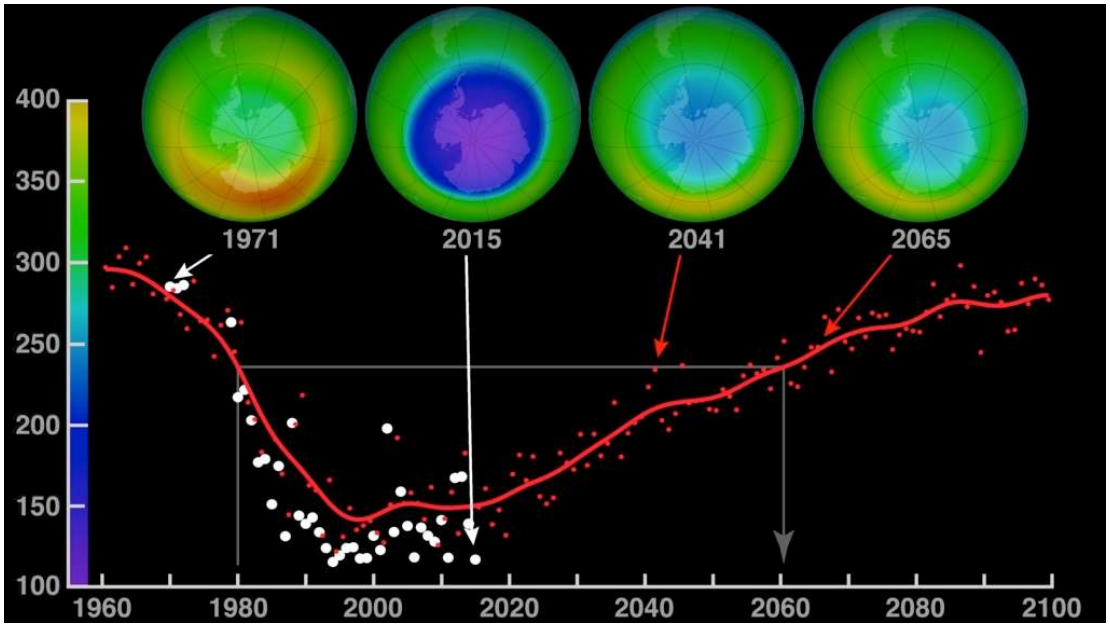


- Examples:





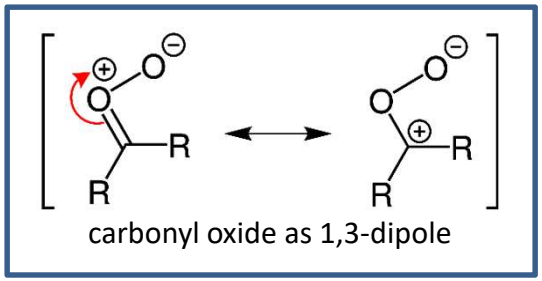


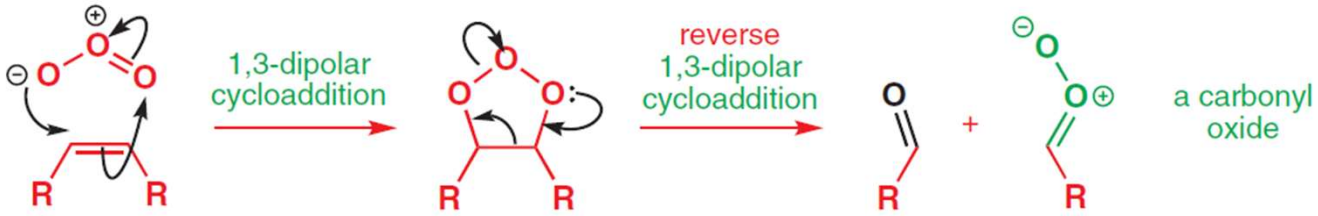


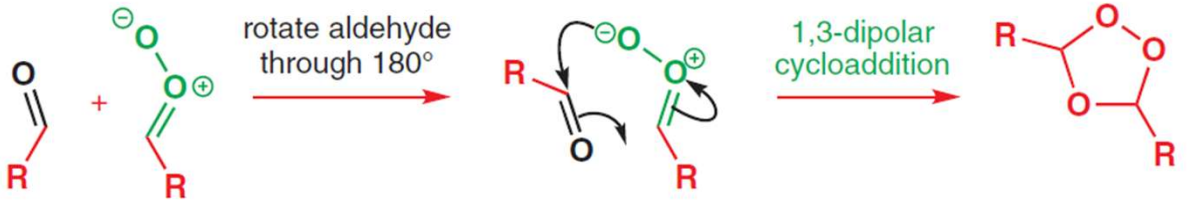
- Ozone can be destroyed by a number of free radical, leading to ozone depletion.
- Ozone hole results in reduction in UV absorption.
- The effects of increased surface UV radiation on human health are a major concern.
- The Montreal Protocol (1987) was designed to protect the ozone layer by phasing out substances (e.g., CFCs chlorofluorocarbons) that are responsible for ozone depletion.

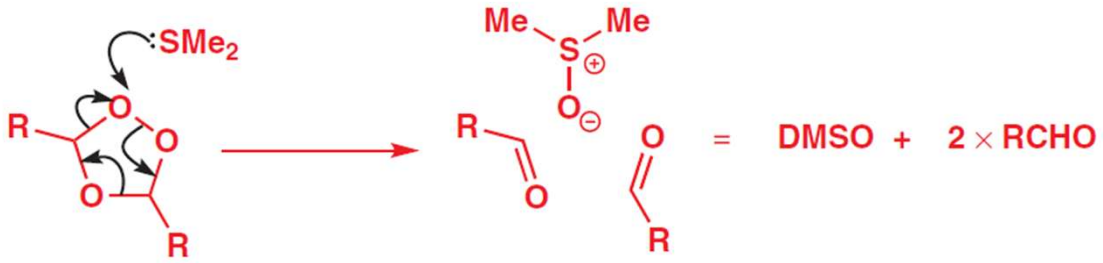
Ozonolysis mechanism







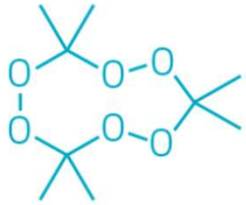




Peroxide compounds

Posted By **Jyllian Kemsley** on Feb 15, 2017 in **Academia**, **Accidents**, **Featured** | **5 comments**

Last week, the Safety Zone reported that **a University of Bristol student had unintentionally made approximately 40 g of triacetone triperoxide (TATP)**, prompting building evacuations and a controlled detonation by an explosives team. Here's a statement about the incident, prepared by **Timothy C. Gallagher**, a chemistry professor and dean of the Faculty of Science, and **Nicholas C. Norman**, head of the school of chemistry.



Triacetone triperoxide



By Aqil Haziq Mahmud
@AqilHaziqCNA

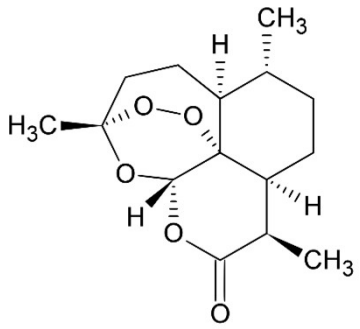
27 Jan 2021 05:00PM
(Updated: 28 Jan 2021 08:32AM)

Singapore

16-year-old Singaporean detained under ISA after planning to attack Muslims at 2 mosques

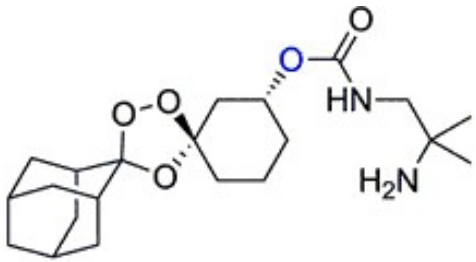
The teen also explored making a triacetone triperoxide (TATP) bomb, and setting fire to the mosques using gasoline, mimicking Tarrant's plan.

Caution: peroxides might be explosive



Artemisinin

2015 Nobel Prize in Physiology or Medicine



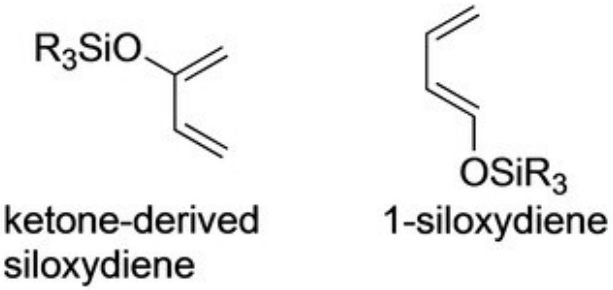
Antimalarial compound containing trioxolane

Cycloaddition reactions

- [4+2] (Diels–Alder)
- [6+4] (triene+diene)
- [4+3] (allyl cation)
- Photochemical [2+2]
- Thermal [2+2]
- [3+2] (1,3-dipolar cycloadditions)

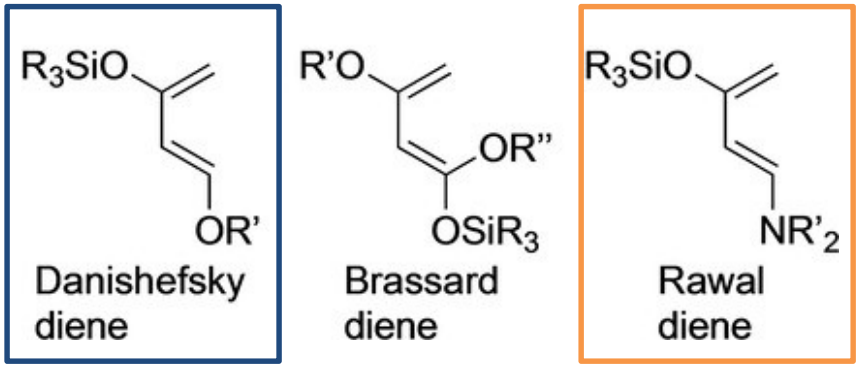
- Feasibility: symmetry of the frontier molecular orbitals (FMOs); Woodward-Hoffmann rules.
- Regioselectivity: bond formation between the atoms with the largest coefficients of the FMOs (HOMO-LUMO for thermal; LUMO-LUMO for photochemical).
- Stereoselectivity: secondary orbital interactions for [4+2] (endo transition state favoured).
- Stereospecificity: within each reactant.

17. Siloxydienes and Aminosiloxydienes



ketone-derived siloxydiene

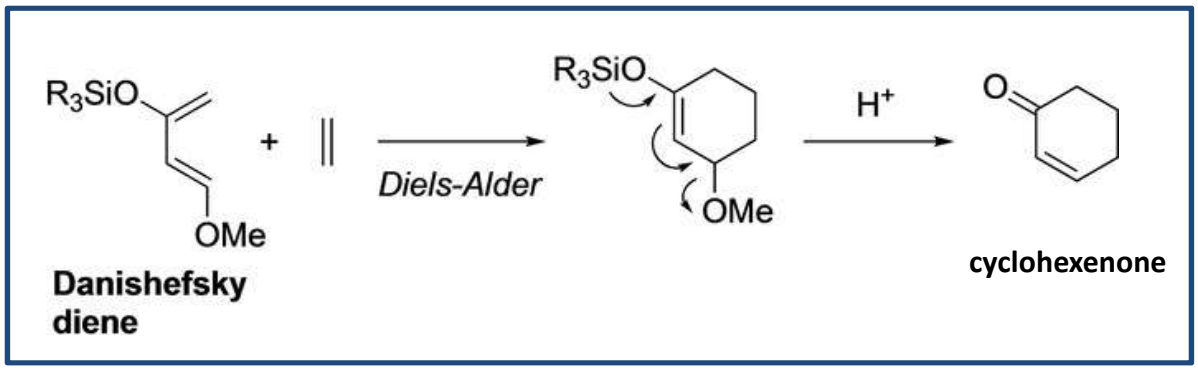
1-siloxydiene



Danishefsky diene

Brassard diene

Rawal diene

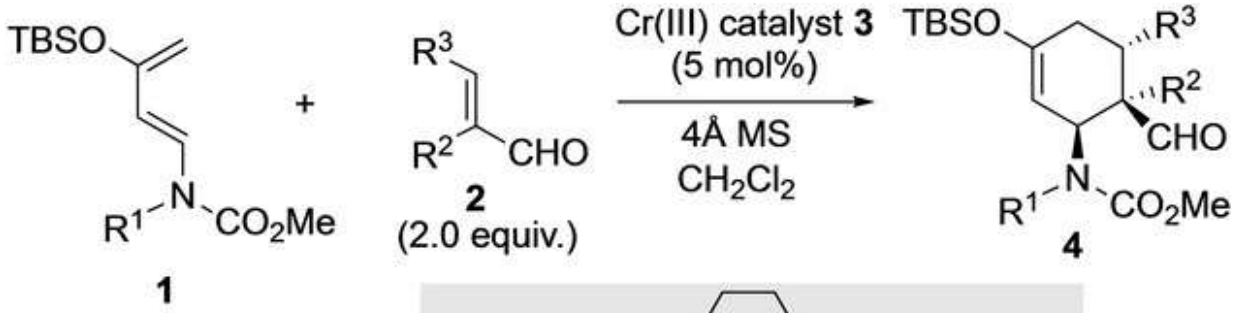


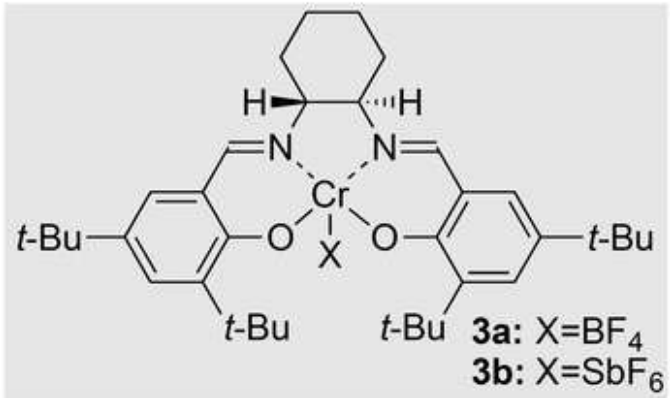
Danishefsky diene

cyclohexenone

Enantioselective Diels-Alder Reaction of Aminosiloxydienes

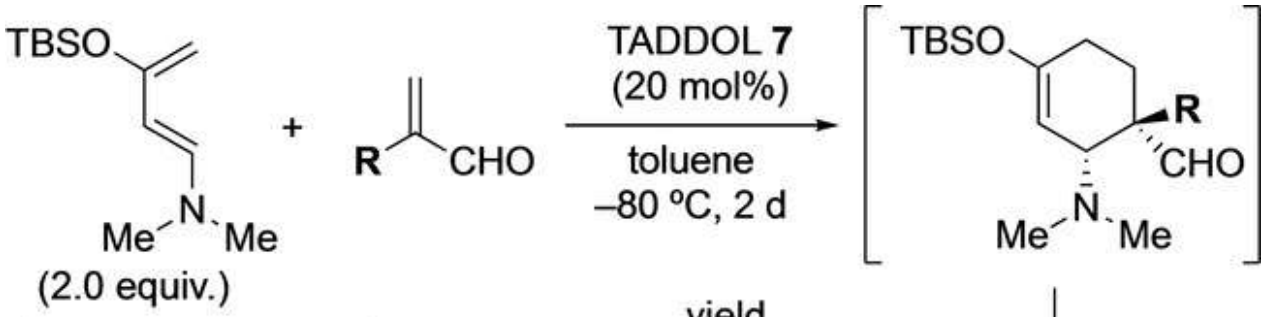
- Highly functionalized cyclohexenes
- Regio- and stereo-selective

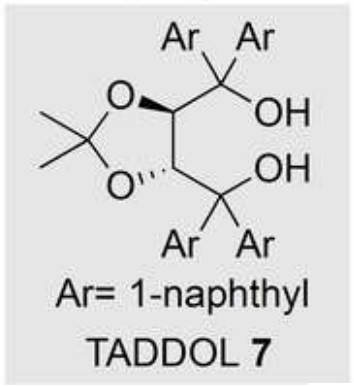




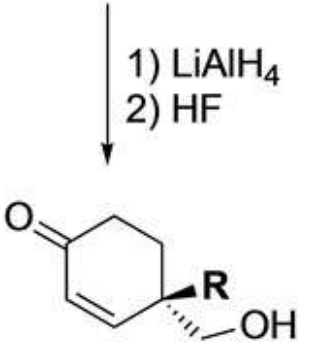
Chiral Lewis acid as catalyst

TADDOL as a hydrogen bond donor catalyst

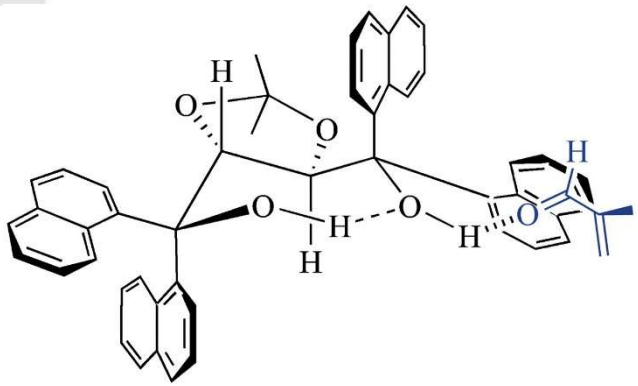




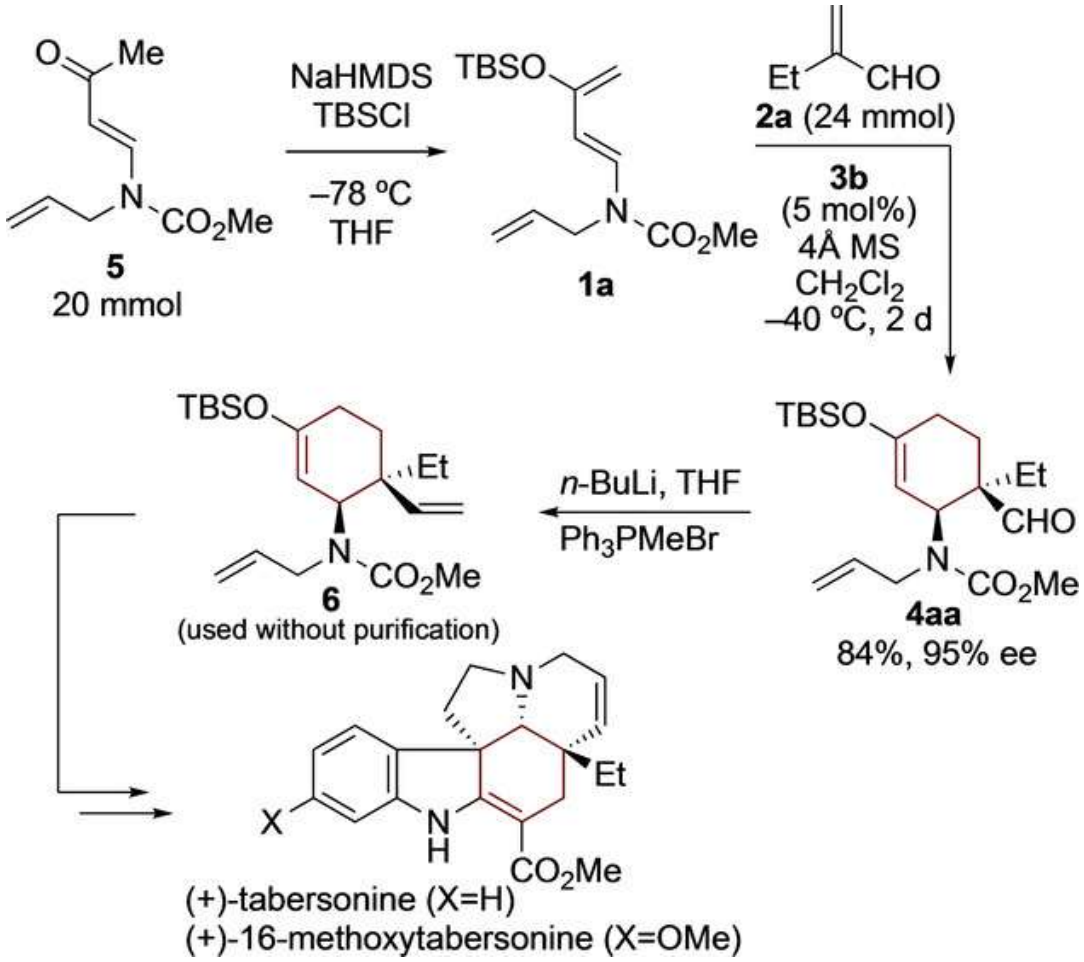
R	yield (%)	% ee
H	77	73
Me	83	91
Et	83	88
*i*Pr	81	92
$PhCH_2$	82	89
$TBSO(CH_2)_2$	80	86



Substituted cyclohexenone



Asymmetric Synthesis of Tabersonines

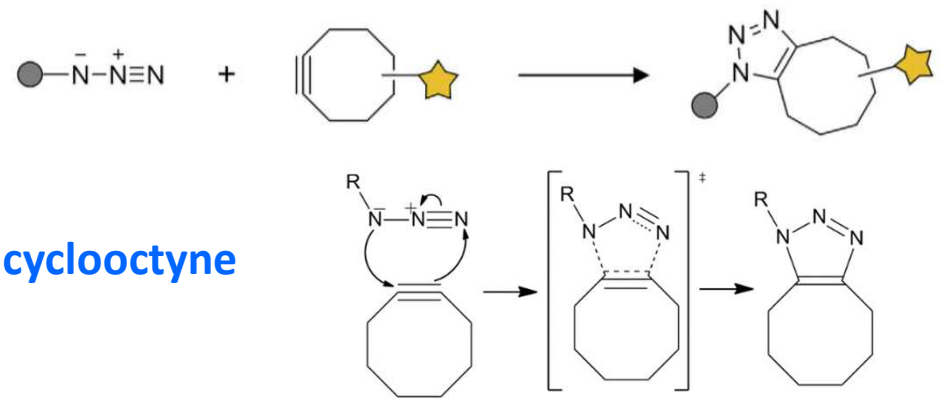


18. Bioconjugate Chemistry

- azide–alkyne cycloaddition



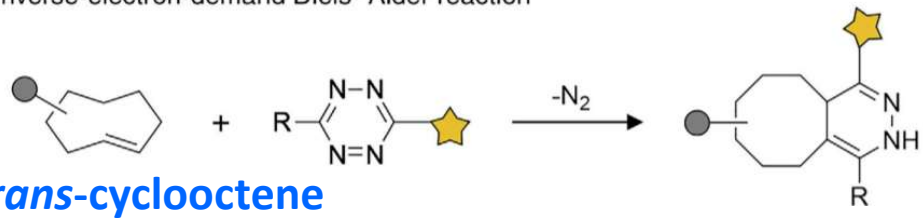
Strain-promoted azide-alkyne cycloaddition



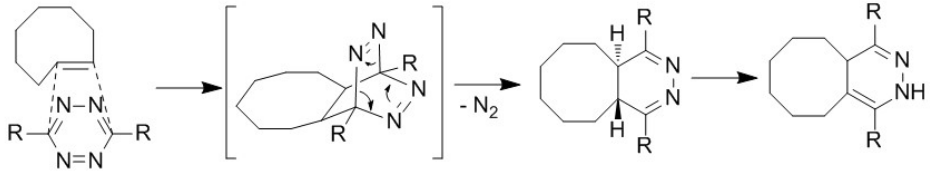
cyclooctyne

- 1,2,4,5-tetrazine–alkene cycloaddition

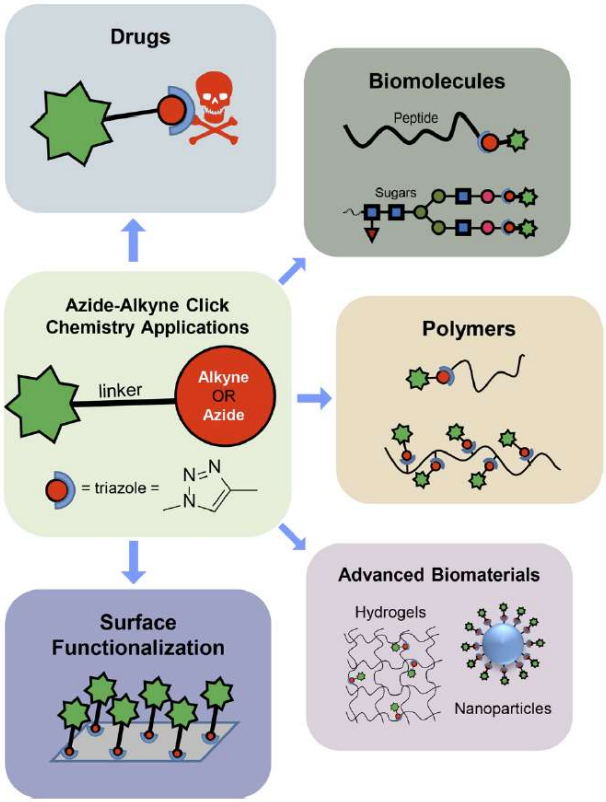
Inverse-electron-demand Diels–Alder reaction



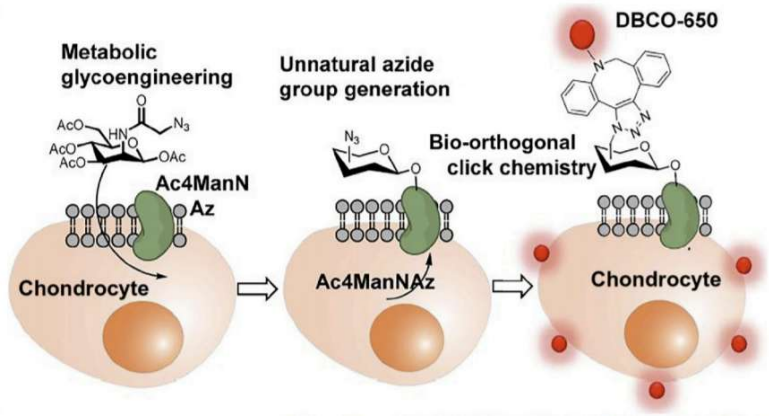
***trans*-cyclooctene**



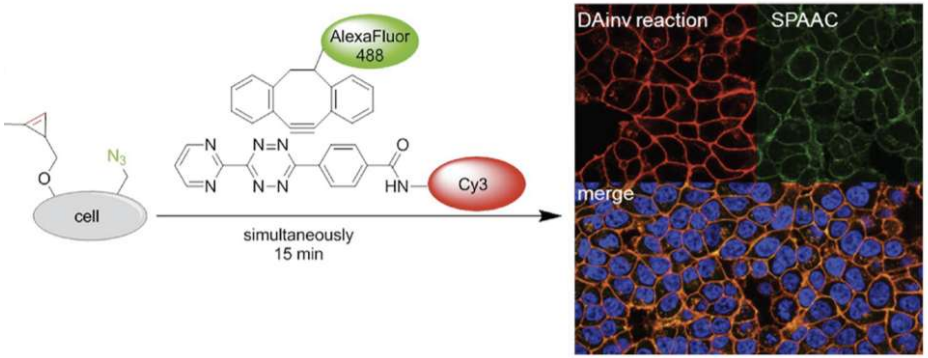
- Selectivity
- Kinetics
- Chemical stability
- Biocompatibility



Bio-imaging

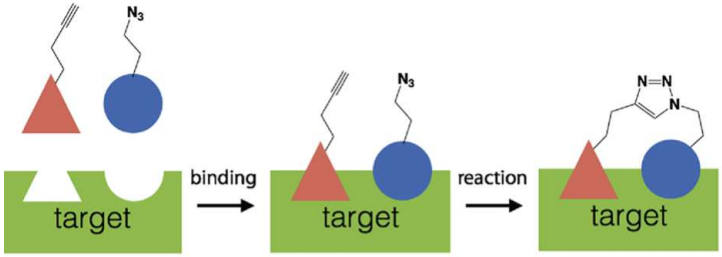


labeling of chondrocytes through metabolic glycoengineering and bio-orthogonal Cufree azide-alkyne click reaction



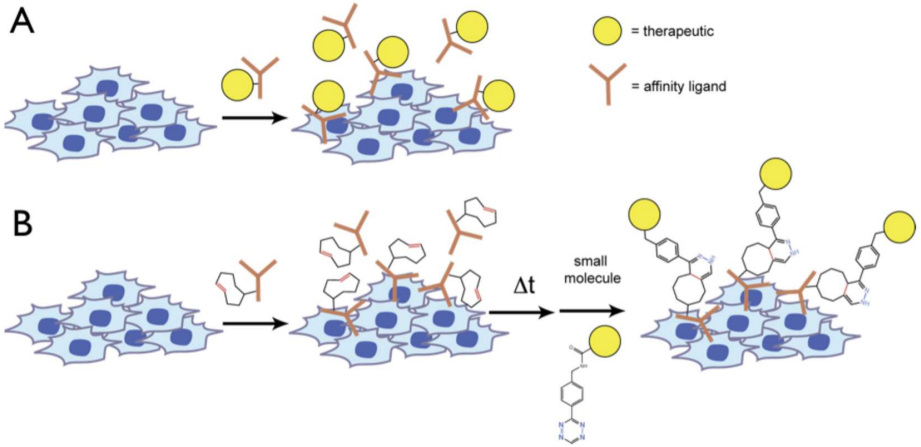
Visualize metabolic incorporation of two different sugars onto cell surface glycans.

Drug discovery



In situ discovery of enzyme inhibitors with enhanced binding affinity

Anticancer therapy



A. Target cells (blue) are exposed to an affinity ligand directly modified by a chemotherapeutic or imaging agent.
B. In the multistep approach, an affinity ligand directly connected to a nontherapeutic bioorthogonally reactive element (trans-cyclooctene) is delivered (reduced toxicity).

Cycloaddition reactions

- A diverse range of ring forming reactions
- Pericyclic mechanism
- Frontier molecular orbitals and Woodward-Hoffmann rules
  - Thermal: total number of electrons (4n+2) allowed
    Special case: thermal [2+2] (suprafacial-antarafacial)
  - Photochemical: total number of electrons (4n) allowed, e.g. [2+2]
- Stereospecific
- Stereoselective
- Regioselective
- Catalysis (Lewis acid, hydrogen bonding interactions)
- Applications
  - Synthesis of functionalized, cyclic structures of various sizes
  - Bio-conjugation